

4.1 (1) $S \rightarrow (L) | a$

(2) $L \rightarrow L, S | S$

① 消除直接左递归 引入 L^-

$L \rightarrow SL^-$

$L^- \rightarrow , SL^- | \epsilon$

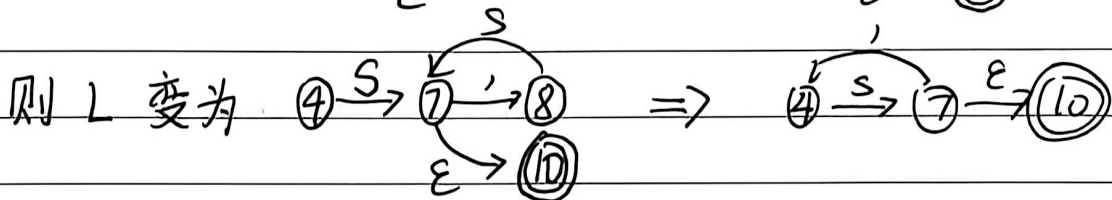
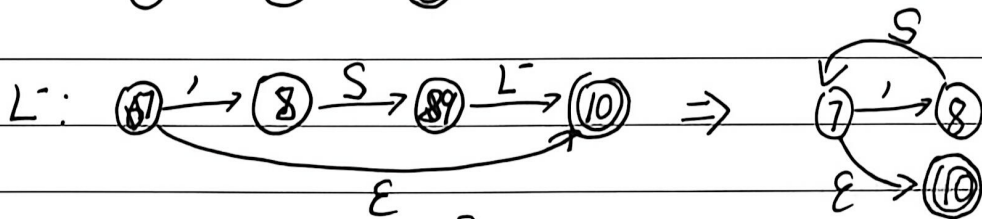
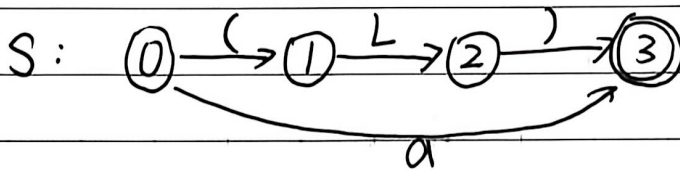
同时易知 S 不存在间接左递归

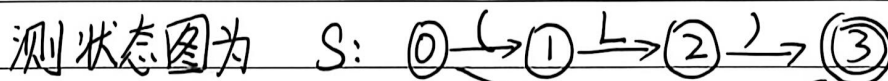
则产生式变为 (1) $S \rightarrow (L) | a$

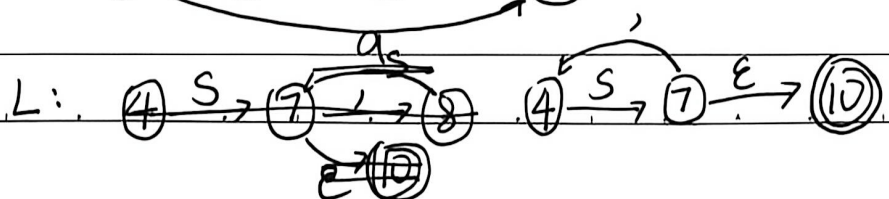
(2) $L \rightarrow SL^-$

(3) $L^- \rightarrow , SL^- | \epsilon$

② 状态转移图

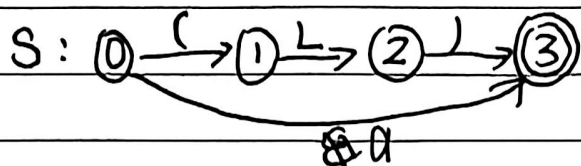


测状态图为 S : 



Date.

③ 预测分析器设计，基于LL(1)进行分析



```
Void S(){
```

```
    if (lookahead == '('){
```

```
        match('(');
```

```
        if (lookahead in FIRST(L)) {
```

```
            LA();
```

```
        } else error();
```

```
        if (lookahead == ')') {
```

```
            match(')');
```

```
        } else error();
```

```
    } else if (lookahead == 'a') {
```

```
        match('a');
```

```
    } else error();
```

```
}
```

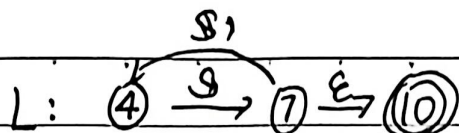
```
Void match(Token tok){
```

```
    if (lookahead == tok){
```

```
        lookahead = scanner.getNextToken();
```

```
    } else error();
```

```
}
```



Void L() {

if (lookahead in FIRST(S)) {

TV(S());

} else error();

While (lookahead == ',') {

match(',',');

if (lookahead in FIRST(S)) {

S();

} else error();

}

if (lookahead in FOLLOW(S)) {

//do nothing

} else error();

补充证明: 该文法满足 LL(1)

$S \rightarrow (L) | a$ $\text{First}(S) = \{ (, a \}$

$L \rightarrow SL^-$ $\text{First}(L) = \text{First}(S) = \{ (, a \}$

$L^- \rightarrow , SL^- | \epsilon$ $\text{First}(L^-) = \{ , , \epsilon \}$

$\text{Follow}(S) = (\text{First}(L^-) - \{ \epsilon \}) \cup \text{Follow}(L^-) \cup \{ \$ \}$
 $= \{ , , \epsilon, \$ \}$

$\text{Follow}(L) = \{) \}$

$\text{Follow}(L^-) = \text{Follow}(L) \cup \text{Follow}(L^-) = \{ \}$

由于 $\text{First}((L)) \cap \text{First}(a) = \{ (\} \cap \{ a \} = \{ (\} \cap \{ a \} = \emptyset$

$\text{First}((, SL^-) \cap \text{First}(\epsilon) = \{ (, \} \cap \{ \epsilon \} = \emptyset$

由于 $\text{First}(\epsilon) = \{ \epsilon \}$, 则又见 $\text{First}((, SL^-) \cap \text{Follow}(L^-)$
 $= \{ (, \} \cap \{ , \} = \emptyset$

所以该文法满足 LL(1)

Date.

(a, (a, a)) 调用序列

↓ S()

match('(')

match L()

S()

| match('a')

| match(',')

S()

| match('(')

| L()

| S()

| match('a')

| match(',')

S()

| match('a')

match(')')

match(')')

4.2 $S \rightarrow aSbS \mid bSaS \mid \epsilon$

不能。原因一：该文法具有二义性(见2.3)，而预测分析器是基于LL(1)文法的。

原因二：可说明文法不满足LL(1)文法的必要条件

$$\text{First}(aSbS) = \{a\}$$

$$\text{First}(bSaS) = \{b\}$$

$$\text{First}(\epsilon) = \{\epsilon\}$$

$$\text{Follow}(S) = \{a, b, \$\}$$

由于 $\epsilon \in \text{First}(\epsilon)$ ，又 $\text{Follow}(S) \cap \text{First}(aSbS) = \{a\}$ 违反条件
所以不满足LL(1)文法。

4.3 $S \rightarrow SS+ \mid SS* \mid a$

$$\text{First}(S) = \{a\}$$

$$\text{Follow}(S) = \text{First}(S) \cup \{+, *\} = \{a, +, *, \$\}$$

因为S是开始符号